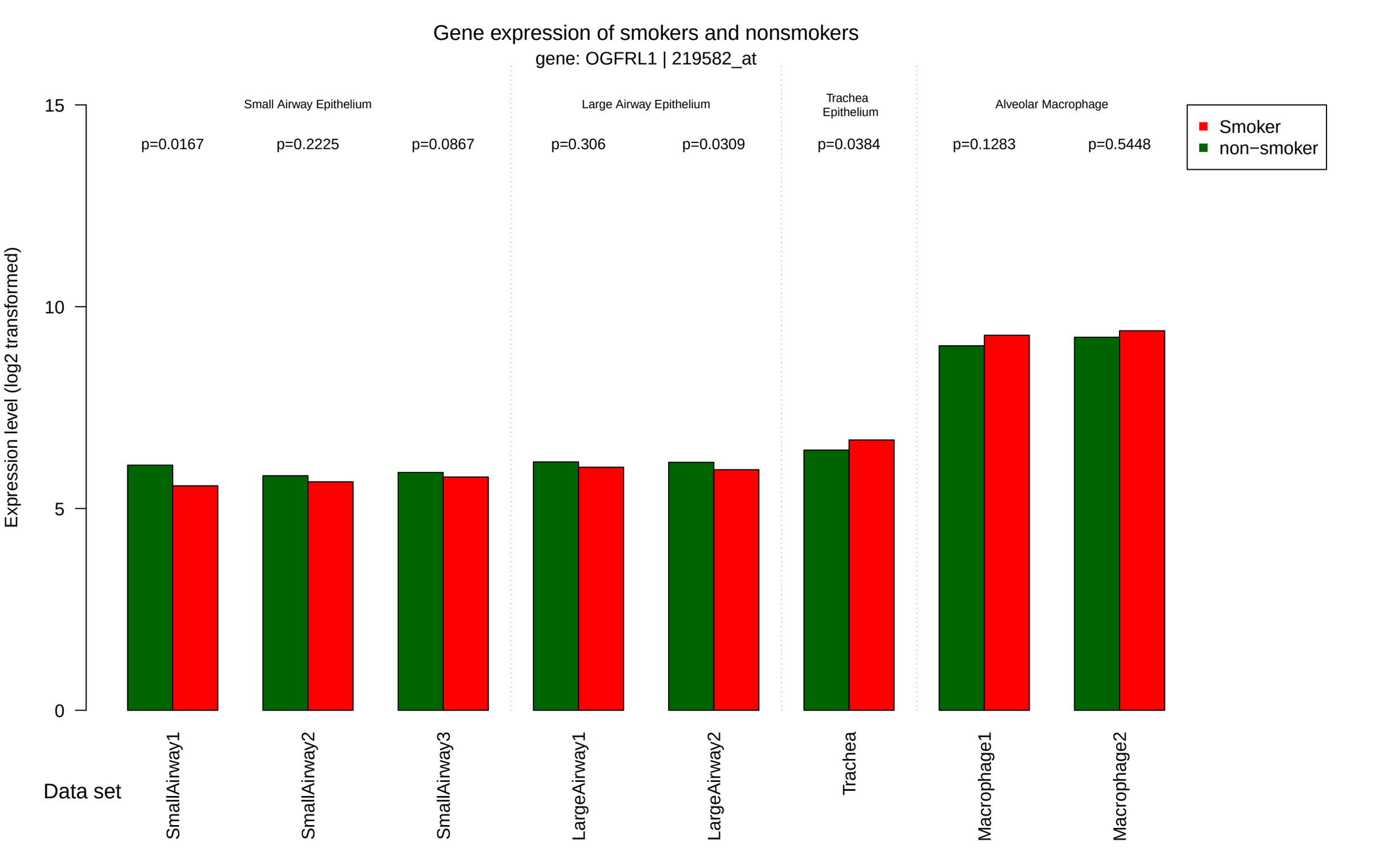
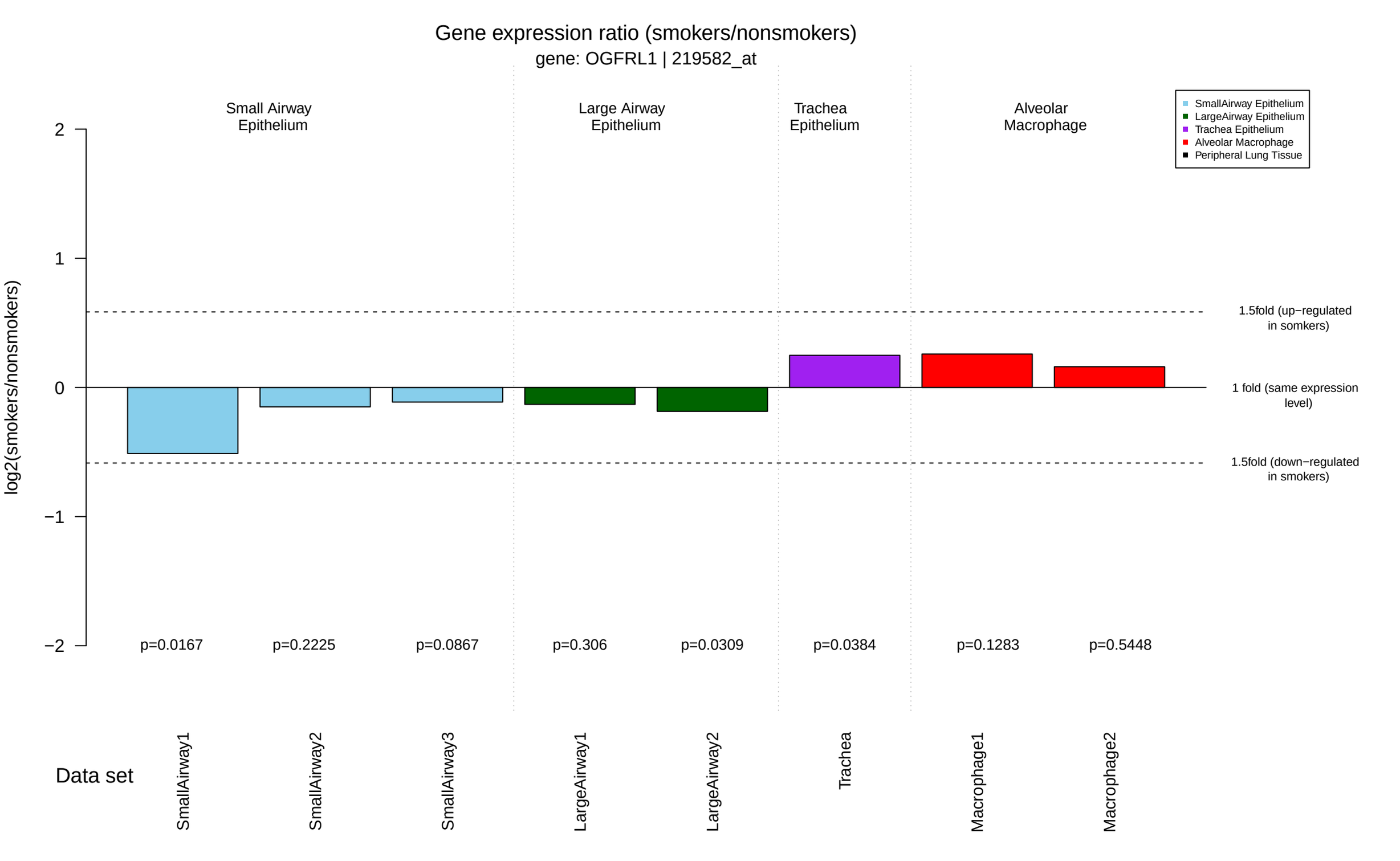
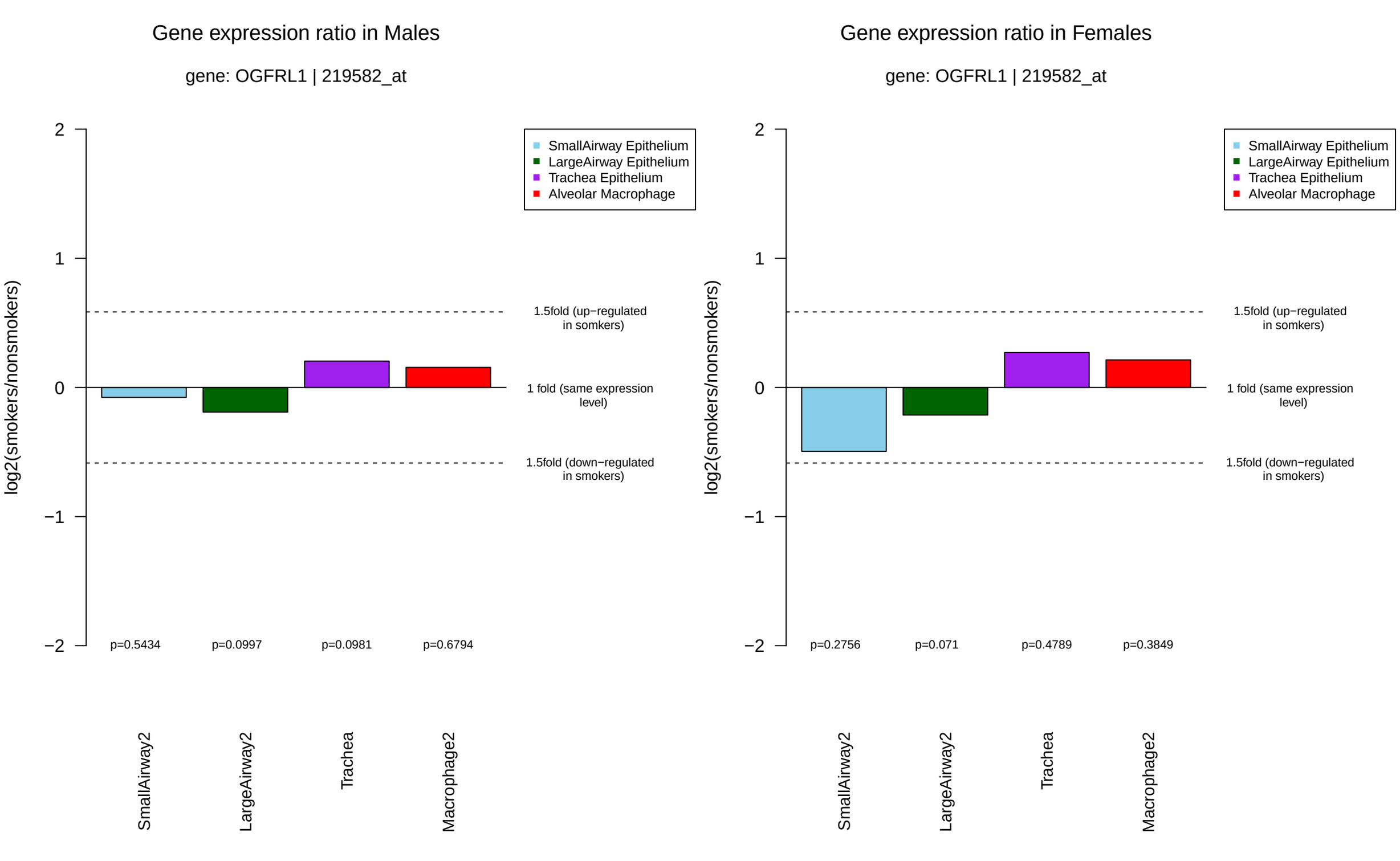
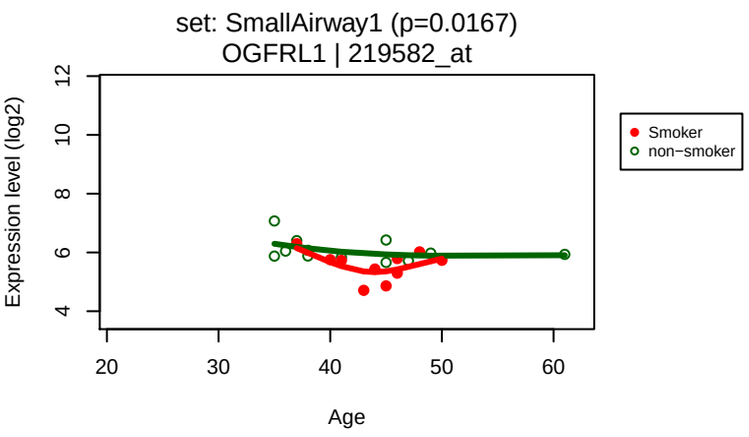
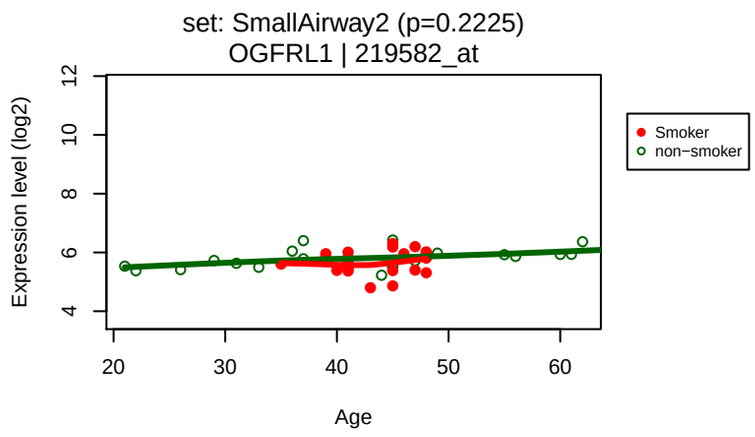
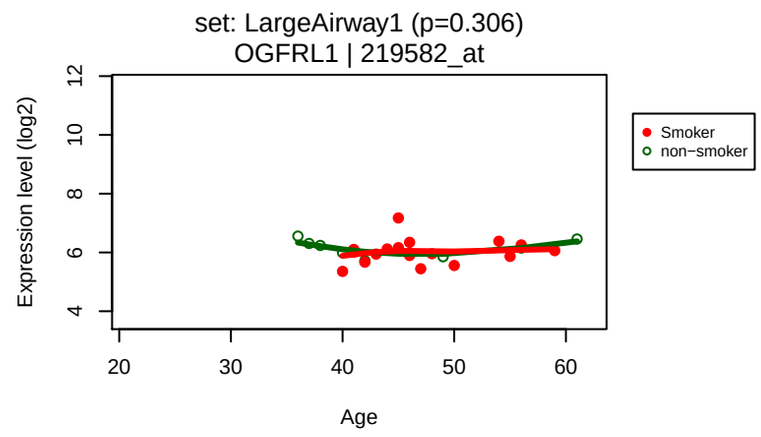

Gene expression of smokers and nonsmokers
gene: OGFRL1 | 219582_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0167
p=0.2225
p=0.0867
p=0.306
p=0.0309
p=0.0384
p=0.1283
p=0.5448
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

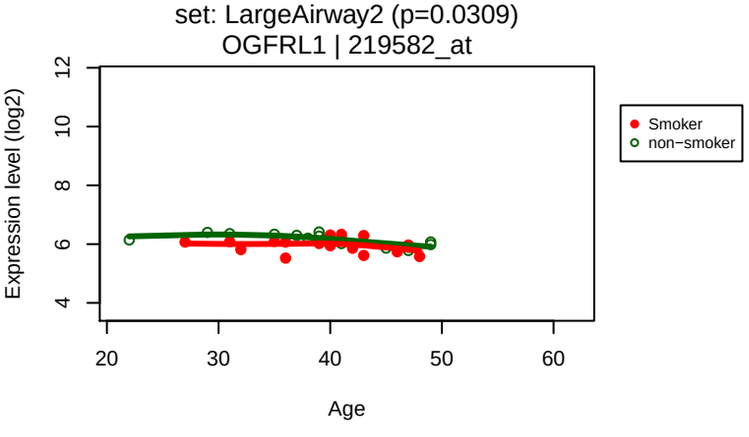

Gene expression ratio (smokers/nonsmokers)
gene: OGFRL1 | 219582_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0167
p=0.2225
p=0.0867
p=0.306
p=0.0309
p=0.0384
p=0.1283
p=0.5448
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

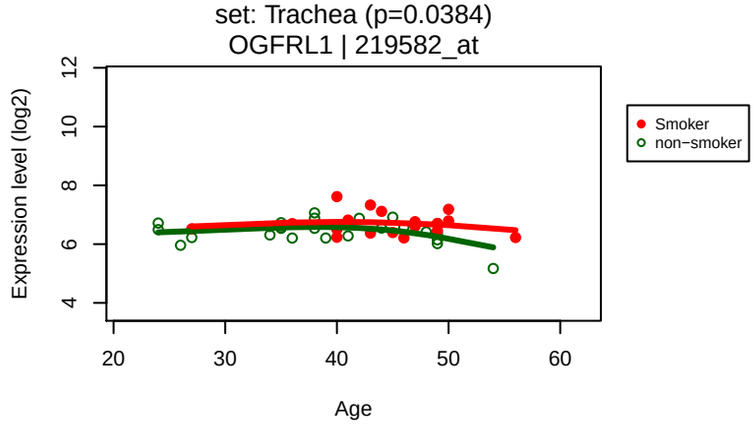

Gene expression ratio in Males
gene: OGFRL1 | 219582_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5434
p=0.0997
p=0.0981
p=0.6794
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OGFRL1 | 219582_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2756
p=0.071
p=0.4789
p=0.3849
SmallAirway2
LargeAirway2
Trachea
Macrophage2

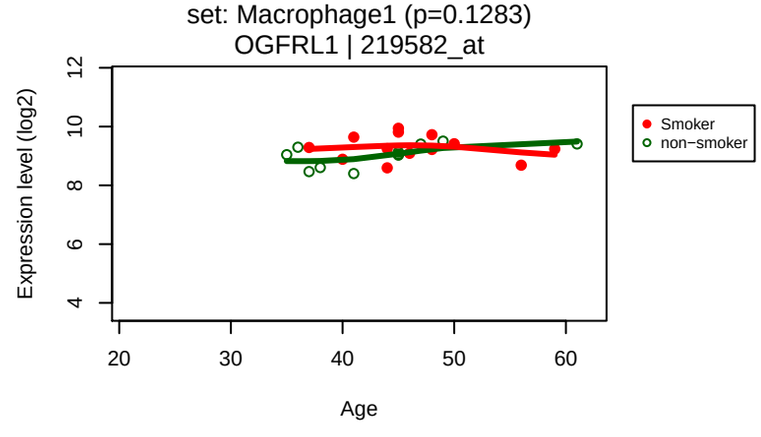

set: SmallAirway1 (p=0.0167)
OGFRL1 | 219582_at
Expression level (log2)
Age
Smoker
non−smoker

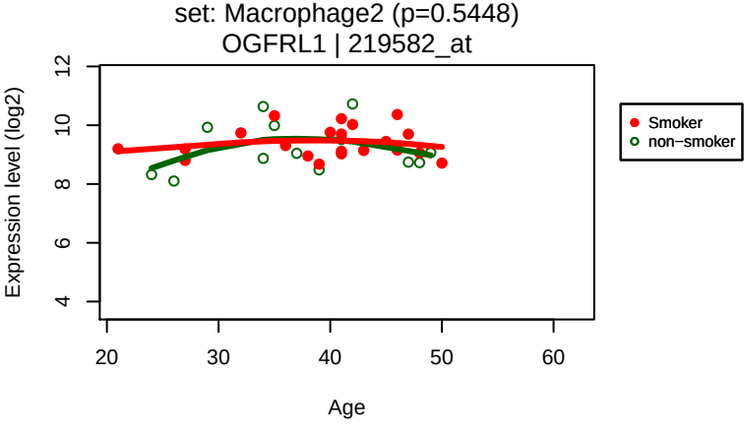

set: SmallAirway2 (p=0.2225)
OGFRL1 | 219582_at
Expression level (log2)
Age
Smoker
non−smoker

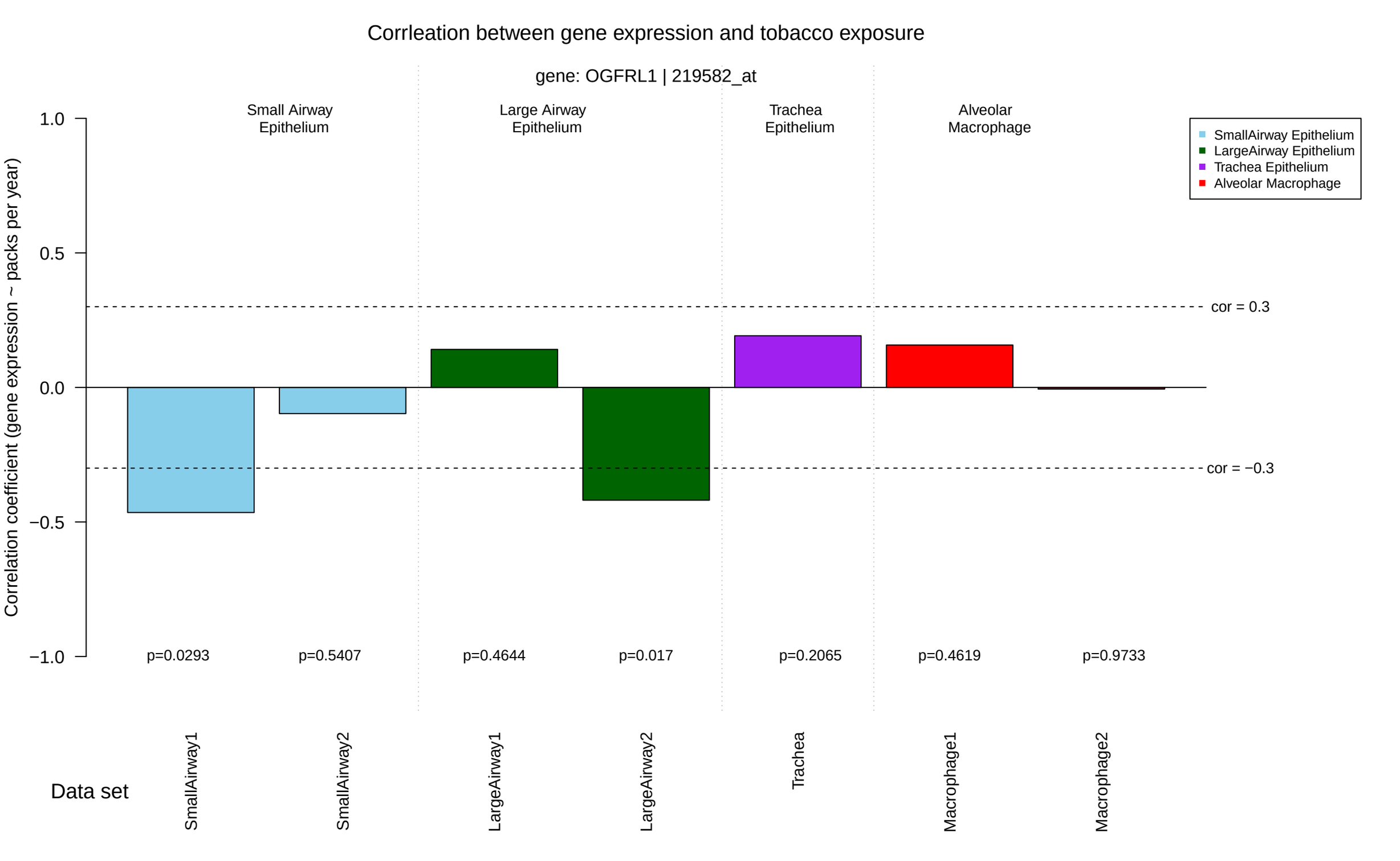

set: LargeAirway1 (p=0.306)
OGFRL1 | 219582_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0309)
OGFRL1 | 219582_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0384)
OGFRL1 | 219582_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1283)
OGFRL1 | 219582_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5448)
OGFRL1 | 219582_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OGFRL1 | 219582_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0293
p=0.5407
p=0.4644
p=0.017
p=0.2065
p=0.4619
p=0.9733
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2